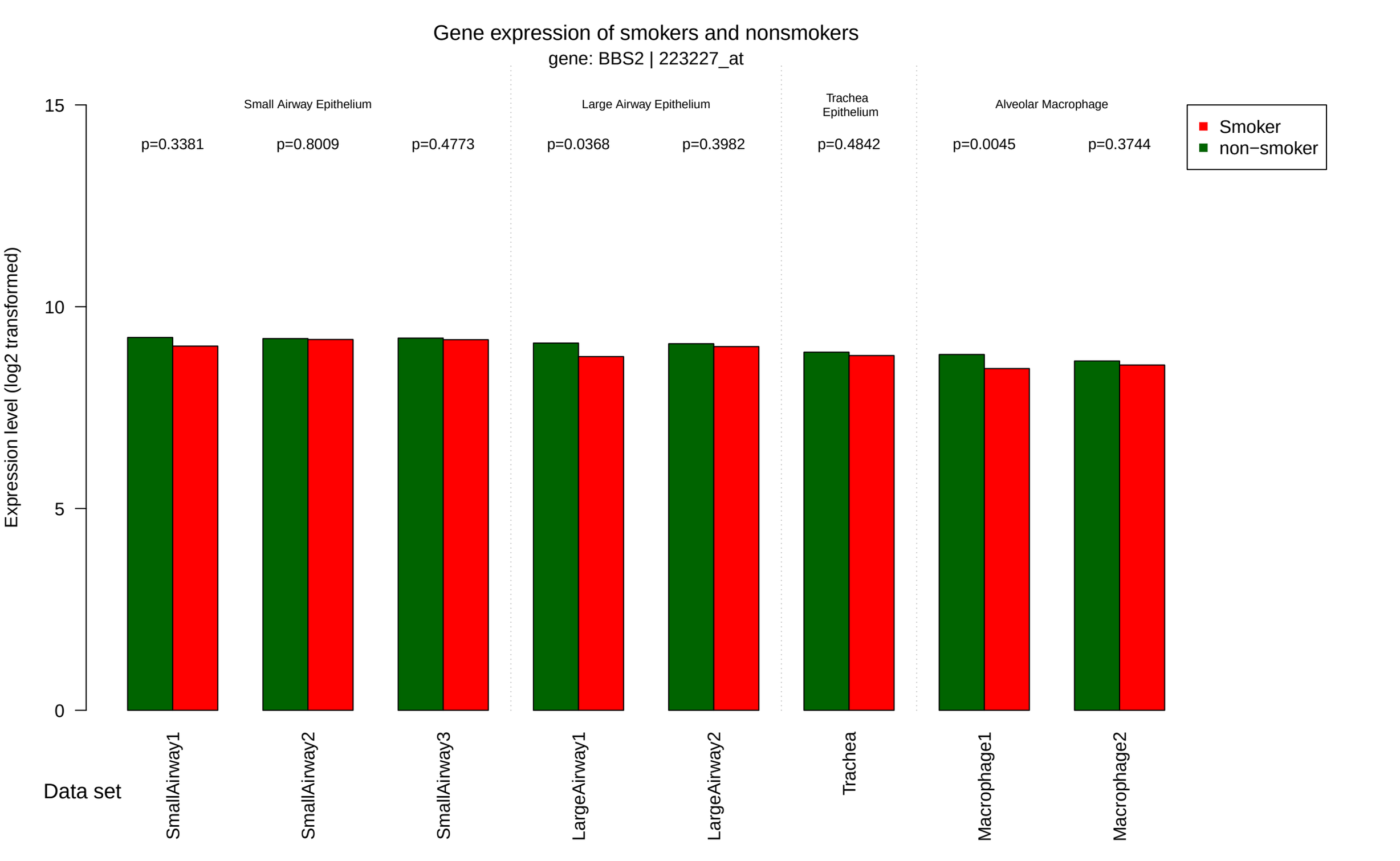
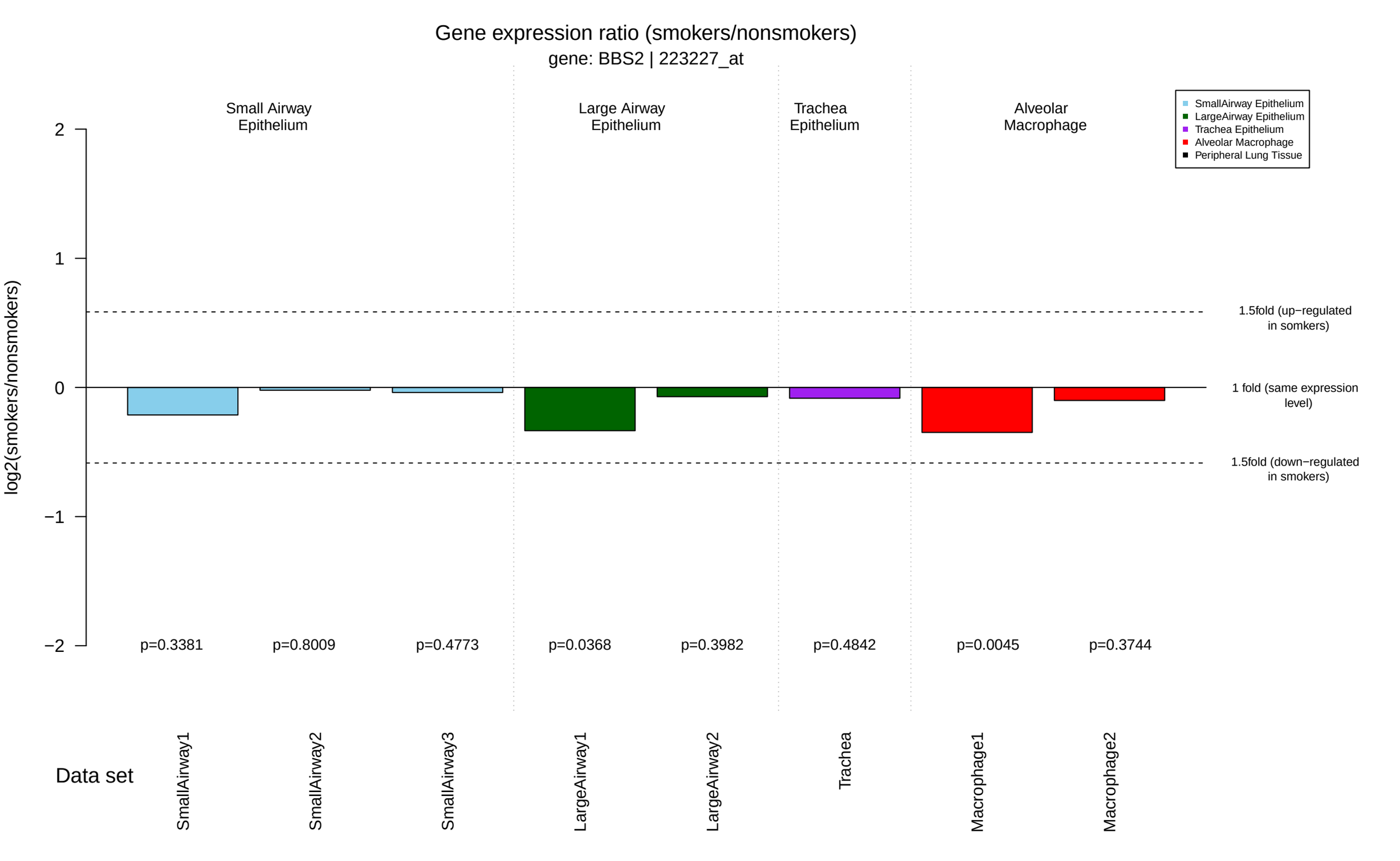
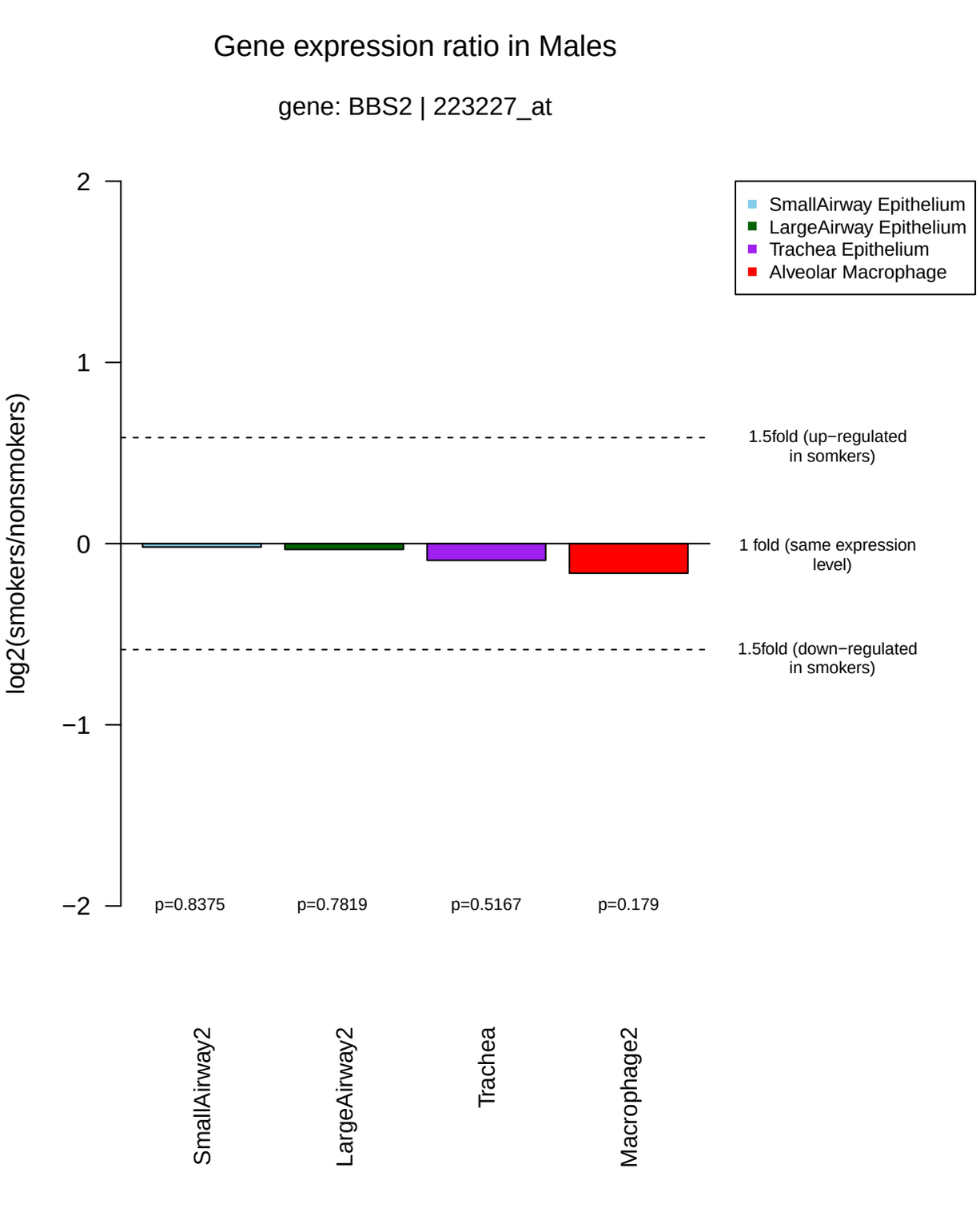
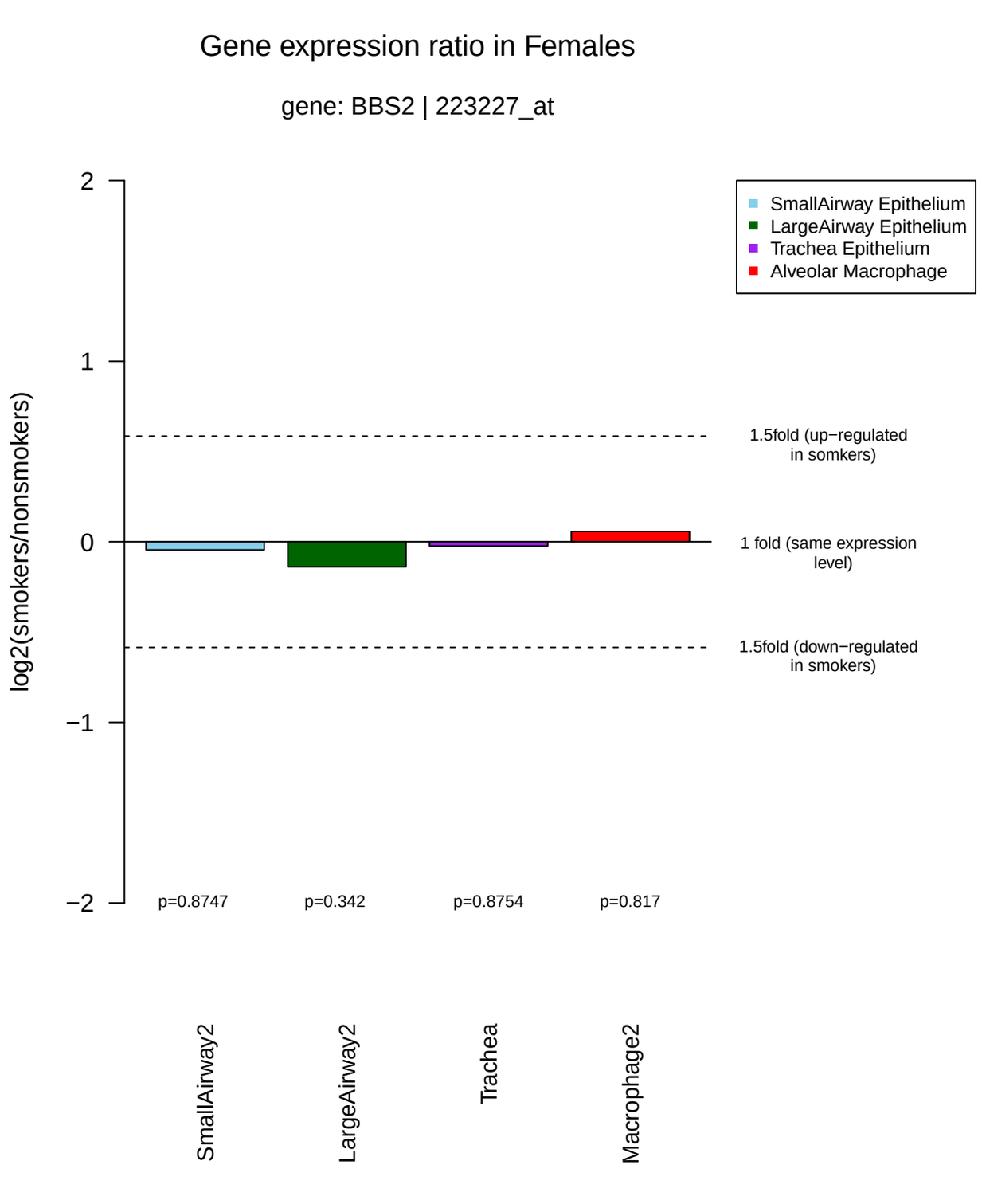
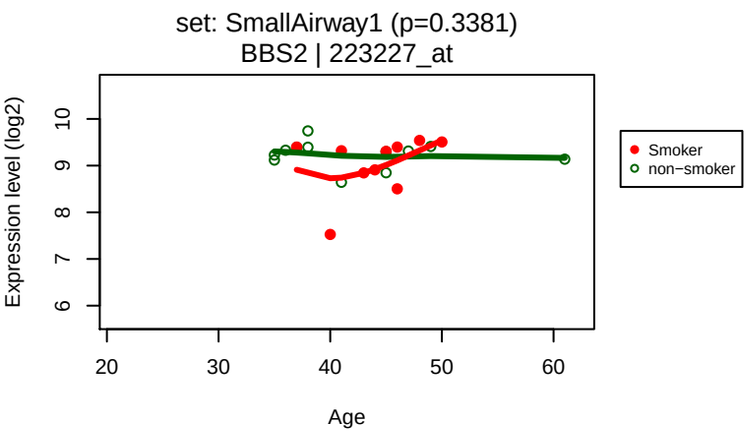
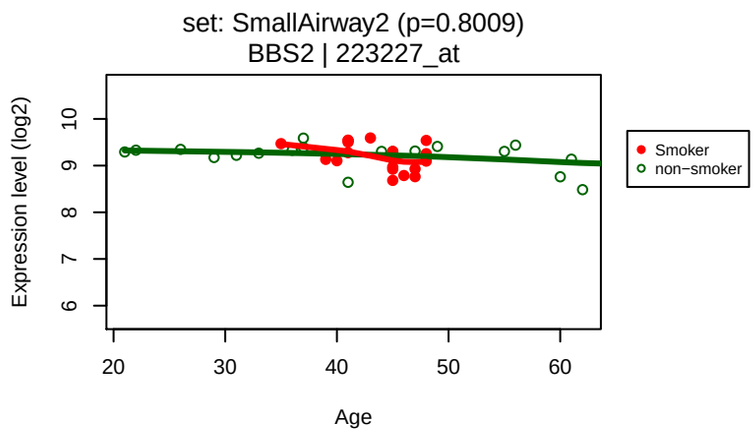

Gene expression of smokers and nonsmokers
gene: BBS2 | 223227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3381
p=0.8009
p=0.4773
p=0.0368
p=0.3982
p=0.4842
p=0.0045
p=0.3744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

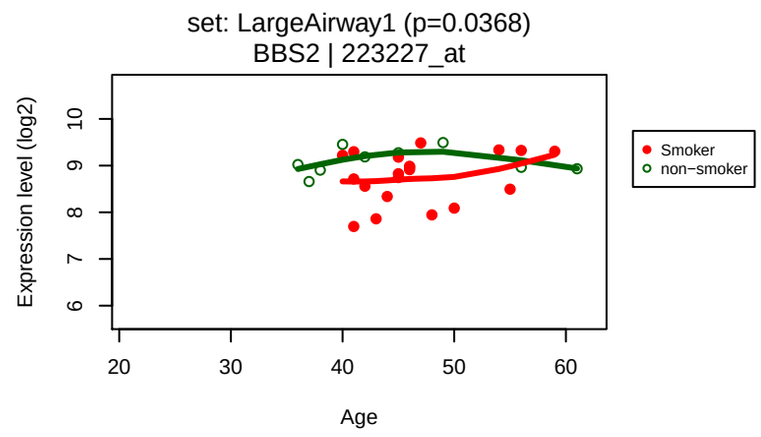

Gene expression ratio (smokers/nonsmokers)
gene: BBS2 | 223227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3381
p=0.8009
p=0.4773
p=0.0368
p=0.3982
p=0.4842
p=0.0045
p=0.3744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

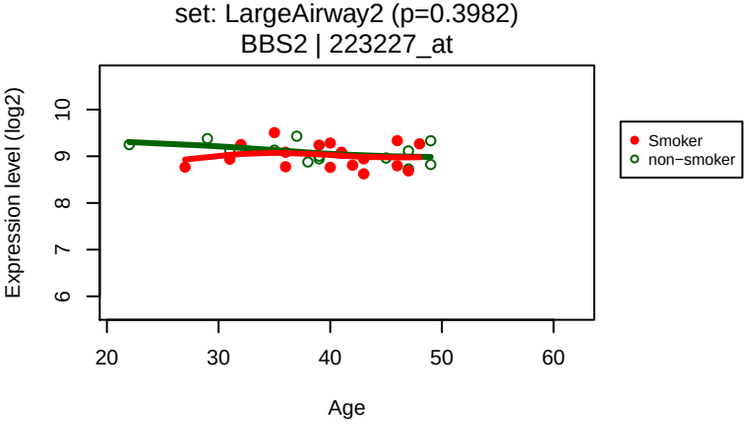

Gene expression ratio in Males
gene: BBS2 | 223227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8375
p=0.7819
p=0.5167
p=0.179
SmallAirway2
LargeAirway2
Trachea
Macrophage2

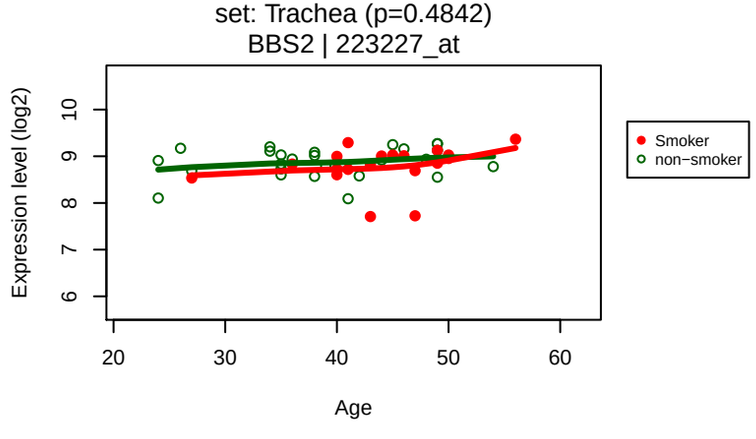

Gene expression ratio in Females
gene: BBS2 | 223227_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8747
p=0.342
p=0.8754
p=0.817
SmallAirway2
LargeAirway2
Trachea
Macrophage2

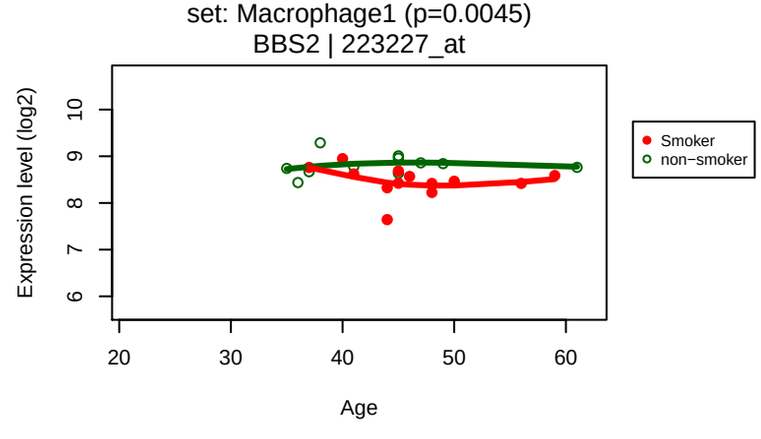

set: SmallAirway1 (p=0.3381)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker

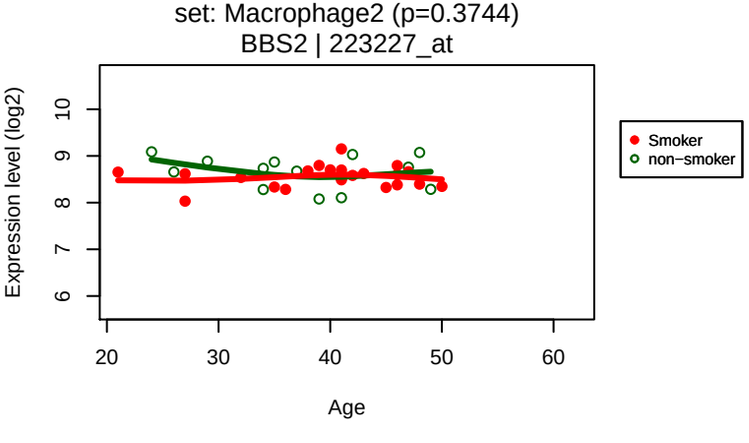

set: SmallAirway2 (p=0.8009)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker

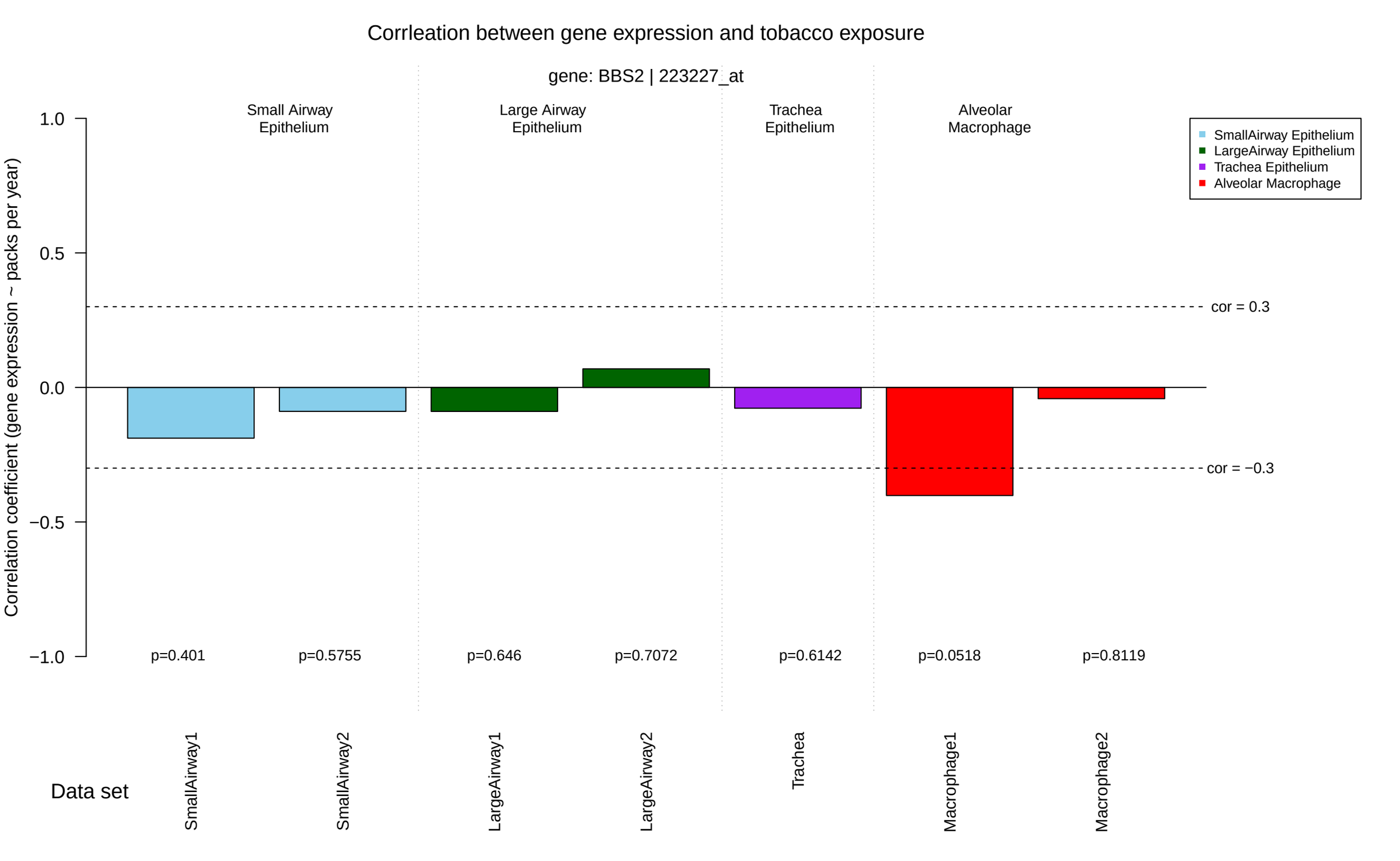

set: LargeAirway1 (p=0.0368)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3982)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4842)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0045)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3744)
BBS2 | 223227_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BBS2 | 223227_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.401
p=0.5755
p=0.646
p=0.7072
p=0.6142
p=0.0518
p=0.8119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2